

COMET CONTACT



TEAM MEMBERS

Isioma Emordi

Shaikh Hamid

Nicole Maing

Adrian Alvarez

Anirudh Vayalali

Ethan Varghese

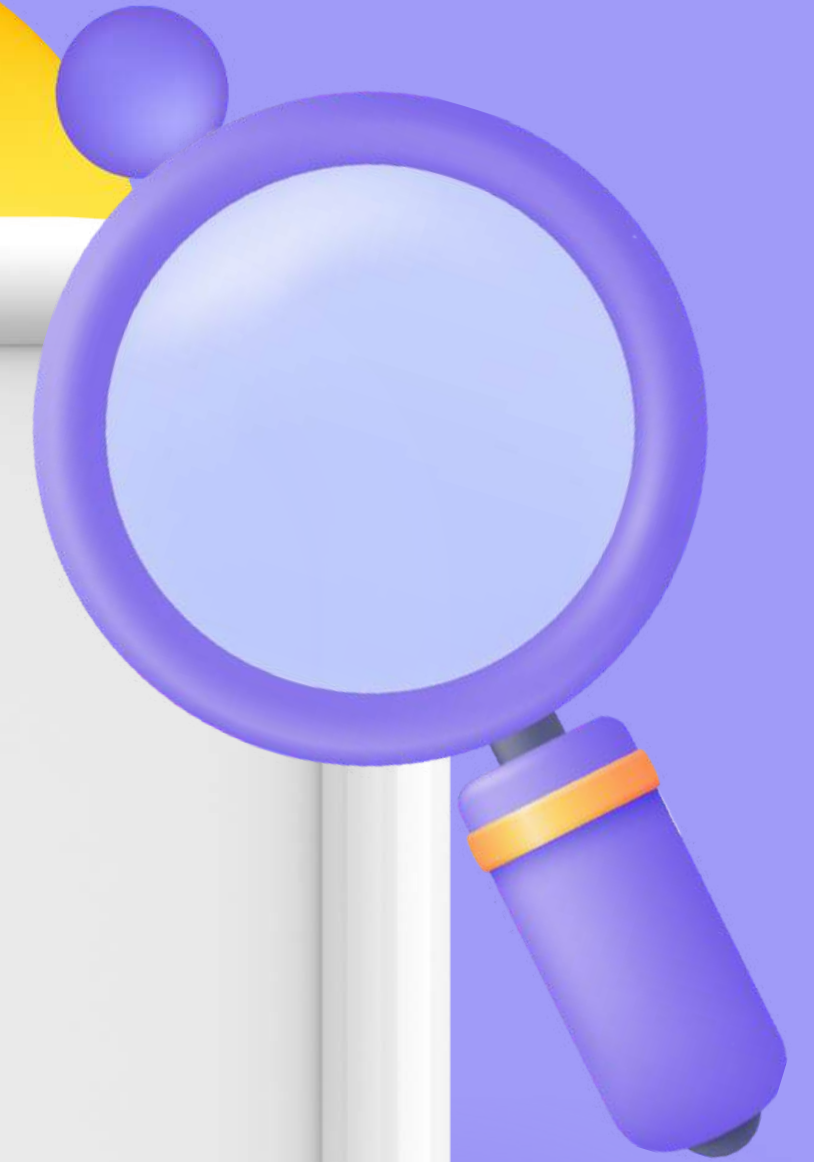
Joshua Gambrell

OBJECTIVE

For organising class project groups, homework groups, study groups, etc., the main method used by a lot of students was to find the name of the student(s), go to outlook, find their email(s), and send a message with (usually) a link or other method to have a constant method of contact (e.g. a link to a discord, whatsapp, or instagram group chat). CometContact (implemented into eLearning) will help facilitate and expedite this process by having a university-wide contact management system to easily find classmates, organize groups, and make connections.

COST ESTIMATION

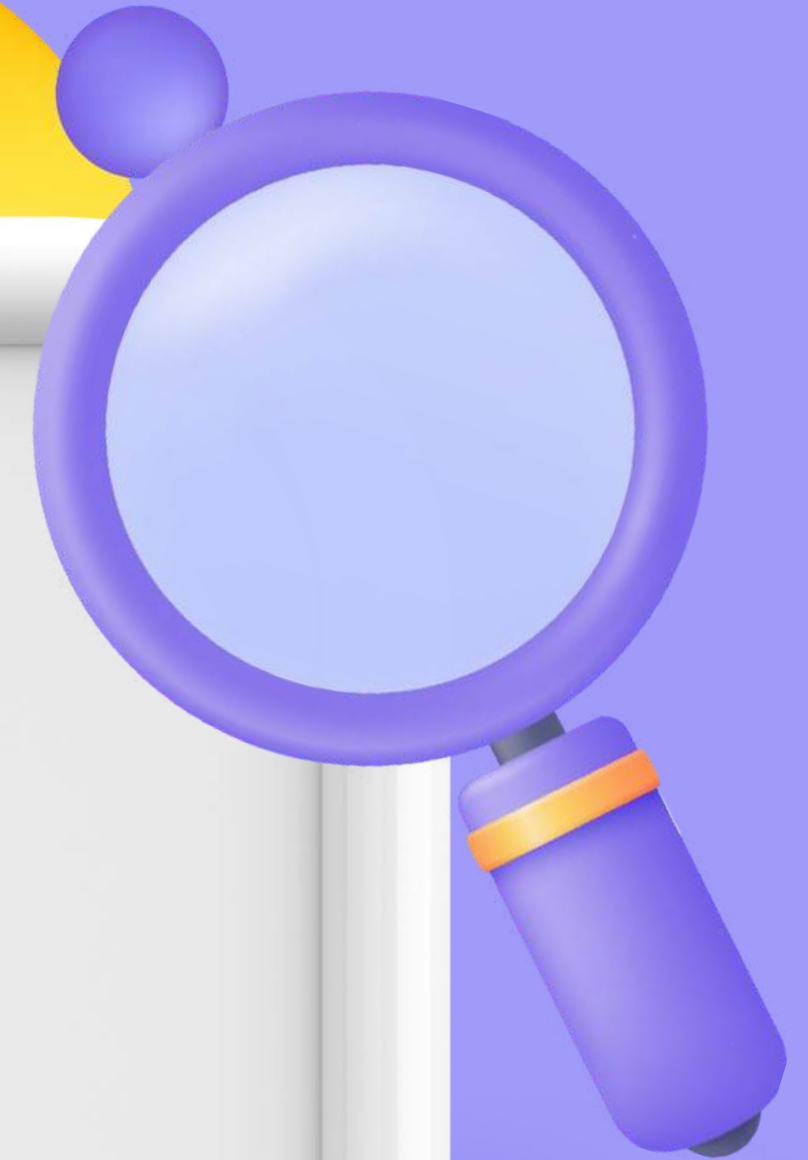
- Estimation Method: **Function Point**
- Total Cost: \$185,228
 - Hardware: \$600
 - AWS Servers
 - Software: \$628
 - Free open-source frame works
 - University integrations
 - GitHub Actions
 - Jira (Project Management)
 - Personnel: \$185,000
 - 6 Full-Time Software Engineers



PROJECT TIMELINE

Assumptions

- Incremental Process Development Model
 - Using Agile Methodology (SCRUM), 2-Week Sprints
- Full-Time Software Engineering Team from UTD's OIT
 - 1 Manager, 1 UI/UX Engineer, 1-2 Backend Engineers, 1-2 Frontend Engineers, 1 QA/Test Engineer
- 8 Hour Workdays (No weekends!)
- APIs, Databases, and Technologies exist from UTD
 - Connecting to eLearning, retrieving students' classes and peers from a database, etc.



THE TIMELINE

Sprint 1: Jan. 1st - Jan. 14th

- Focus on Requirement Refinement
- Database Design
- UTD SSO (NetID Authentication) (UTD APIs)
- GitHub Repository Setup

Jan.1

Sprint 3: Jan. 29th - Feb. 11th

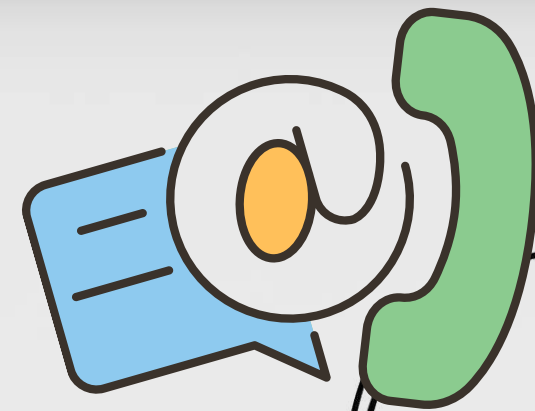
- Contact Groups Feature
 - Backend: API for Create/Delete Groups
- Search Feature
- Blacklist Contacts Feature
- Testing!

Sprint 2: Jan. 15th - Jan. 28th

- Backend: API for Add/View/Edit/Delete Contacts
- Frontend: Basic Dashboard and Contact List UI
- Manual Contact Creation Feature
- Testing!

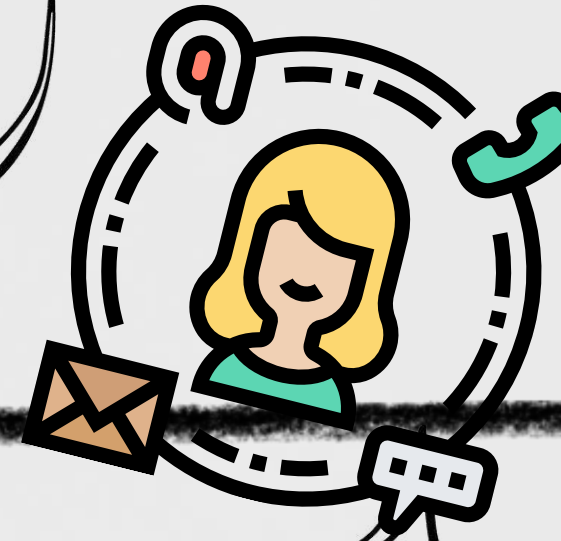
Sprint 4: Feb. 12th - Feb 25th

- Direct Call and SMS from Contacts
- Support Multiple Phones and Social Accounts
- Sorting Feature (By Name or Group)
- Testing!



Sprint 6: March 11th - March 24th

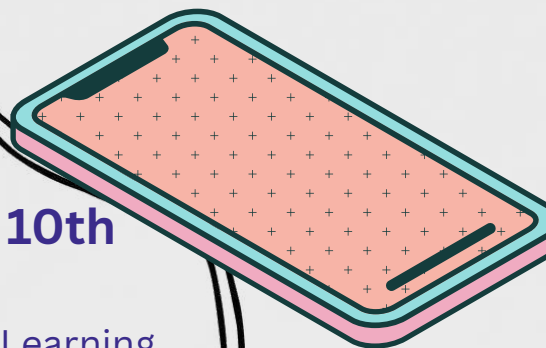
- Import/Export Contact List Feature
- Photo Upload for Contacts Feature
- Frontend: UI Polish and Refinement
- Testing!



**April
5th**

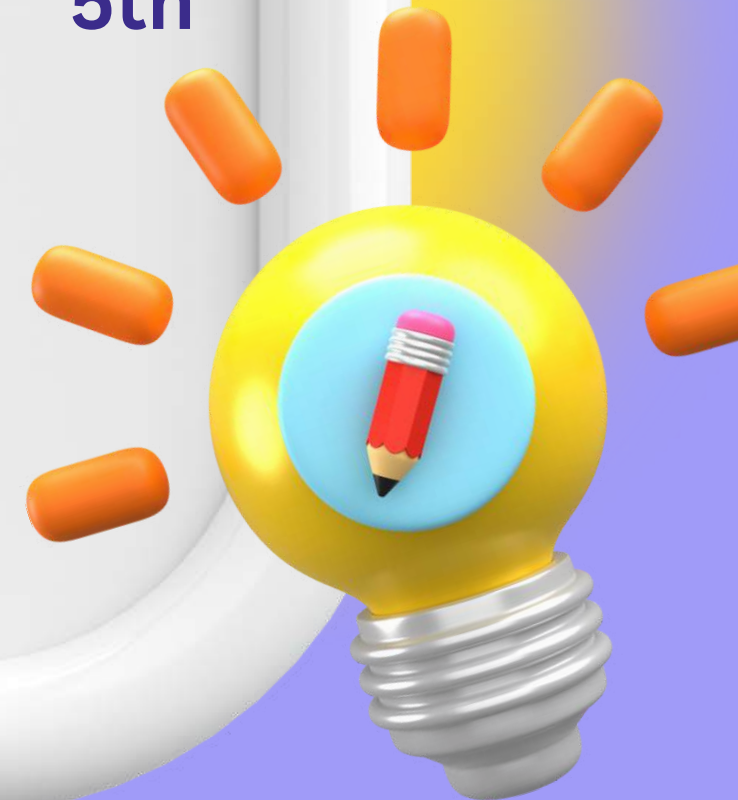
Sprint 5: Feb. 26th - March 10th

- Automated Class List from UTD/eLearning
- Auto-Grouping Based on Classes Feature
- Testing!



Sprint 7: March 25th - April 5th

- Integration Testing
- User Testing
- Bug Fixes
- Deployment!



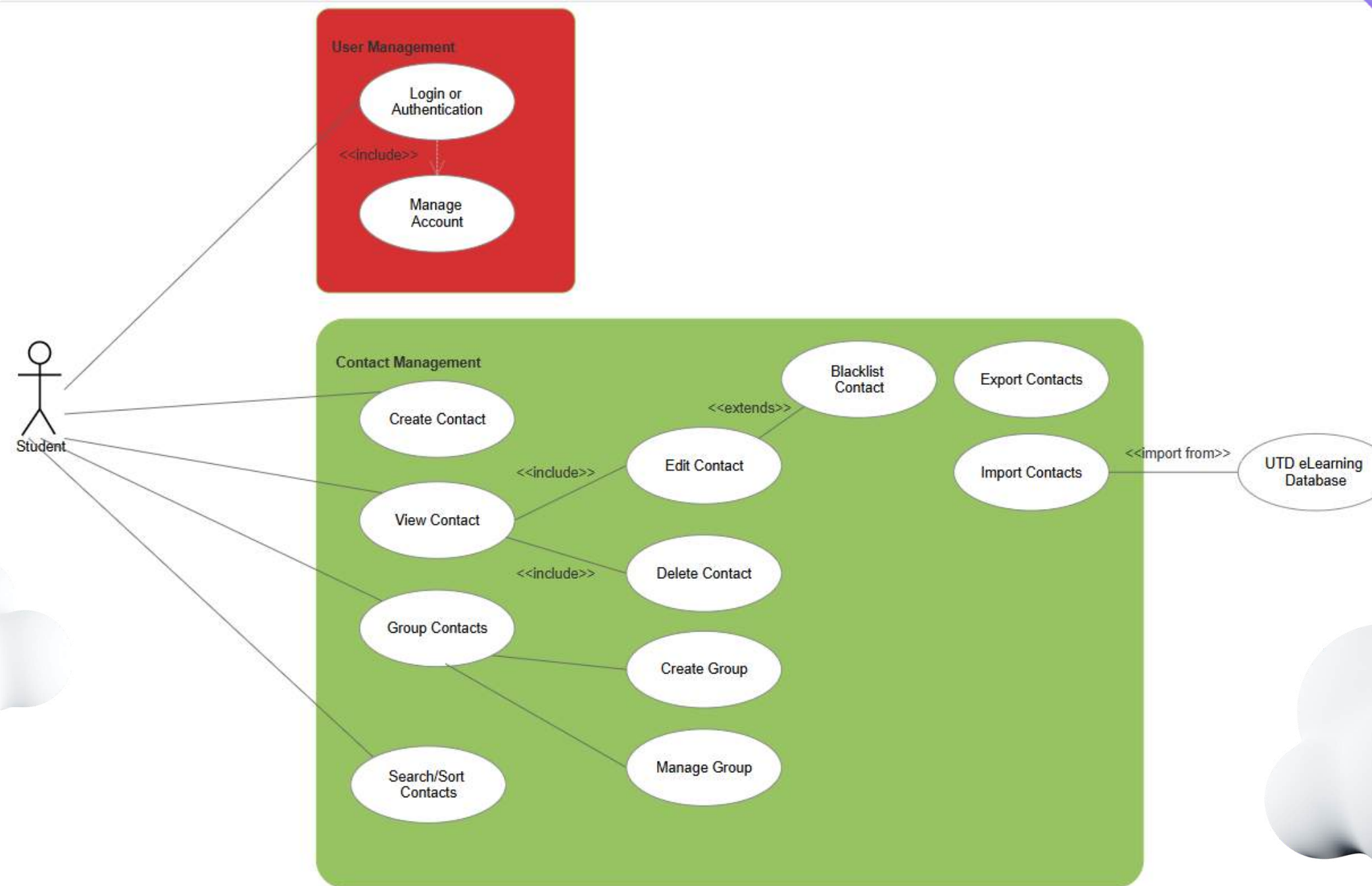
FUNCTIONAL REQUIREMENTS

- Classmate Discovery: Search and filter classmates by course, section, name, and NetID.
- Group Creation & Joining: Create course-specific groups with capacity limits set by group creators.
- Contact Management: Add or delete contacts using the built-in filtering and lookup system.
- Profile Management: Update editable profile fields.
- Blocking & Security Control: Block/unblock contacts to prevent interaction and visibility.
- User Verification & Enrollment Validation: Authenticate through UTD SSO.

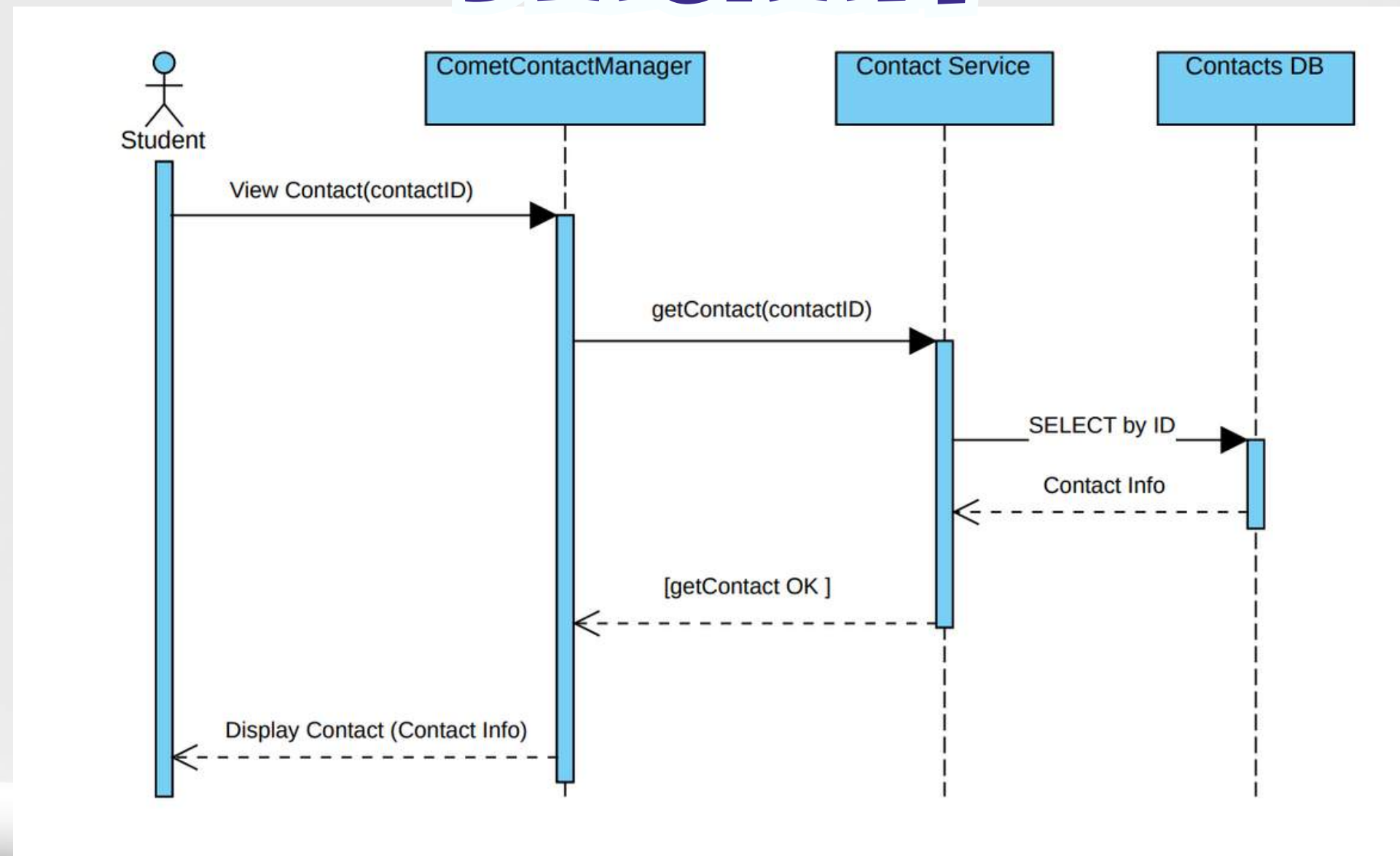
NONFUNCTIONAL REQUIREMENTS

- Performance: Must handle large numbers of students and notifications smoothly.
- Reliability: Must stay available during peak times like enrollment and exams.
- Scalability: Must support growing student populations without issues.
- Maintainability: Codebase must be easy to update and fix.
- Development Process: Must use staged environments (Dev/Test/Prod).
- Security & Standards: Must follow university security and coding standards.
- Technology Constraints: Must use university-approved technologies.
- Ethical & Safety: Must protect students with blocking and safety features.
- Legal Compliance (FERPA): Must follow FERPA rules and protect student data.

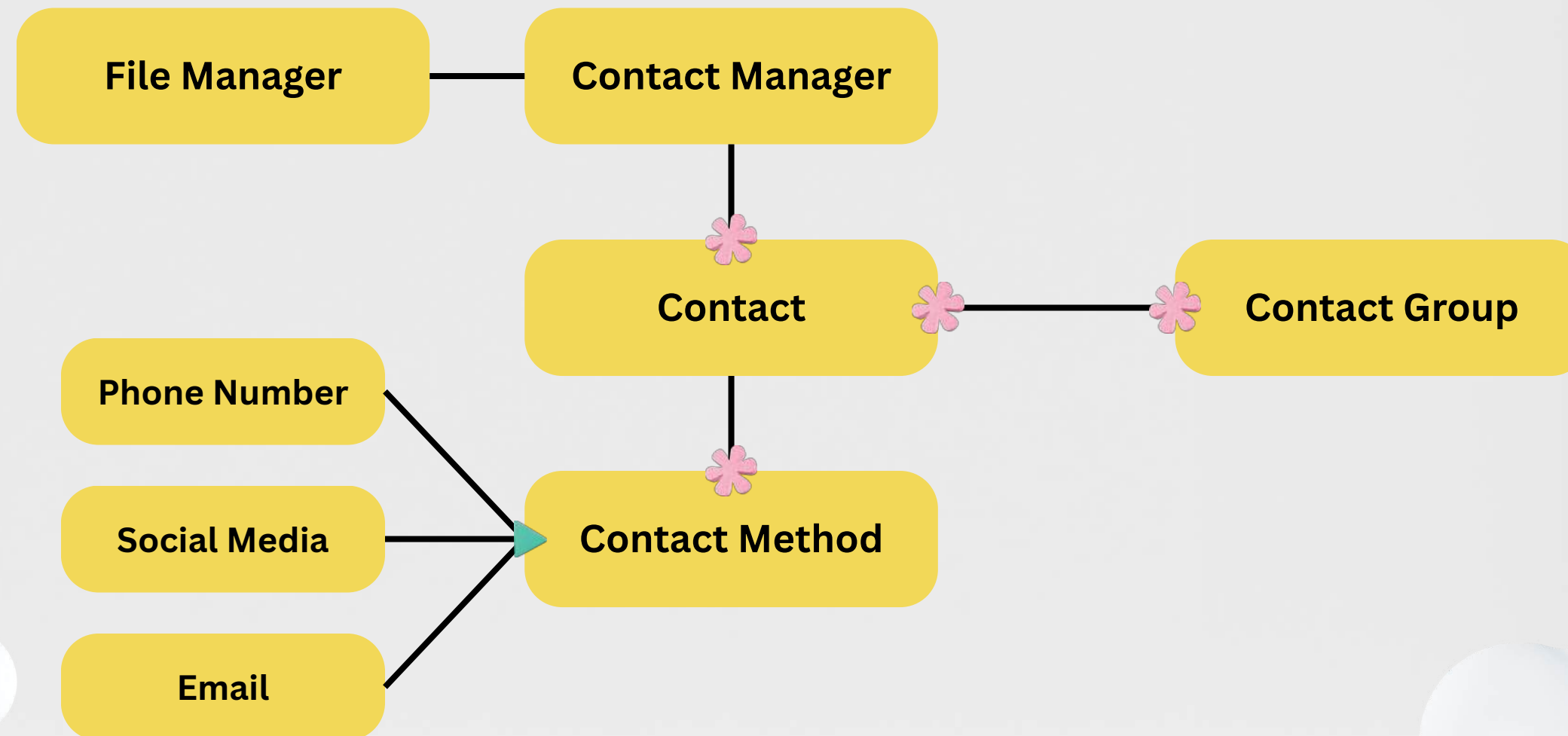
USE CASE DIAGRAM



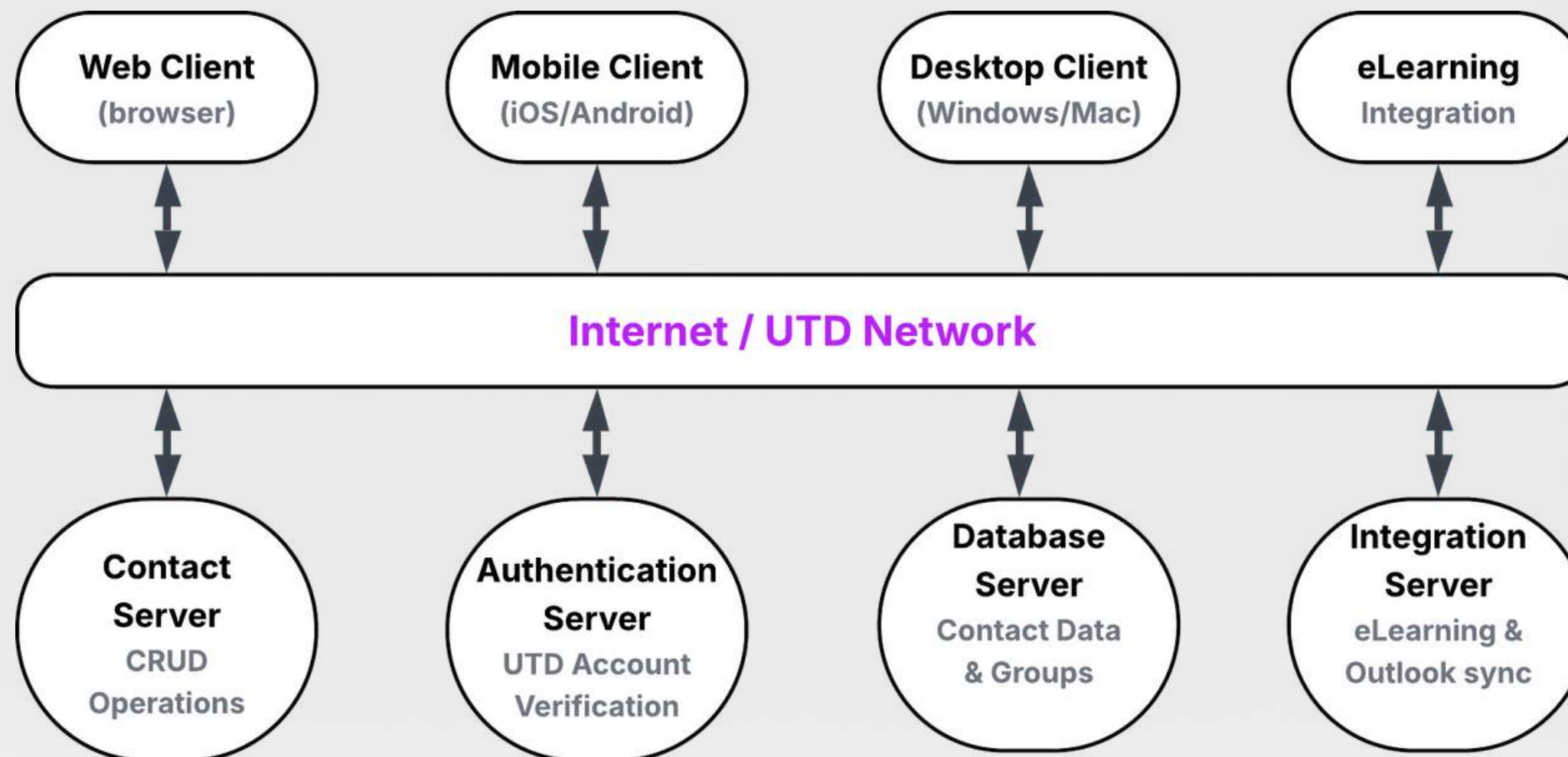
SEQUENCE DIAGRAM



CLASS DIAGRAM



ARCHITECTURAL DESIGN



**THANK
YOU**

